

PHASE 7: Project Documentation

FitTrack – AI: Intelligent Fitness Tracking & Personalized Health Insights

Project Overview: This phase records the technical and user-facing information needed to understand, install, run, test, and maintain the FitTrack – AI backend application.

1. Project Overview

FitTrack – AI is a RESTful backend for activity tracking, user health profile management, workout log recording, and AI-generated fitness insights and workout recommendations.

2. Technology Documentation

Technology	Purpose
Node.js	Runs the backend application.
Express.js	Provides HTTP routing and middleware.
MongoDB	Stores users, fitness metrics, and activity logs.
Mongoose	Defines schemas and communicates with MongoDB.
JWT	Authenticates protected API requests.
bcryptjs	Hashes and compares user passwords.
Fitness Data API	Provides activity tracking metrics when configured.
Google Gemini	Generates fitness summaries and workout recommendations when configured.
Postman	Tests the REST API endpoints.

3. Environment Variables

Variable	Purpose
PORT	Server port; project defaults to 5000.
MONGO_URI	MongoDB connection string.
JWT_SECRET	Secret used for JWT signing and verification.
FITNESS_API_KEY	Optional key for live activity and tracking data.
GEMINI_API_KEY	Optional key for Gemini AI generation.

4. API Documentation

Endpoint	Access	Description
POST /api/auth/register	Public	Register a user and return a JWT.
POST /api/auth/login	Public	Authenticate a user and return a JWT.
GET /api/auth/profile	JWT	Return the logged-in user's profile.
POST /api/activities	JWT	Add a workout log or activity entry.
GET /api/activities	JWT	List the user's logged activity history.
PUT /api/activities/:id	JWT	Update an activity entry.
DELETE /api/activities/:id	JWT	Delete an activity entry.
GET /api/fitness/:type	Public	Return current fitness metrics for an activity type.
POST /api/ai/fitness-summary	JWT	Generate a natural-language fitness summary.
POST /api/ai/fitness-recommendation	JWT	Generate workout and health-based recommendations.

5. Data and Security Notes

- Passwords are not returned as part of the profile response and are stored using `bcrypt` hashing.
- JWT authentication is required for profile, activity, and AI routes.
- Activity log updates and deletes verify that the entry belongs to the authenticated user.
- Configuration secrets are intended to be supplied through environment variables rather than hard-coded into the application.
- The supplied repository contains a `.gitignore` file and an environment configuration file; secret values should not be committed to a public repository.

6. Repository Documentation

The supplied repository contains `README.md` with setup instructions, project structure, API documentation, and Postman testing guidance. The Postman collection is included as `postman_collection.json`.

7. Scope Clarification

Note: Some wording in the original source document refers to posts, categories, comments, analytics, role matrices, and blog workflows. These do not appear in the supplied FitTrack code structure and are therefore excluded from this phase-wise documentation.